

习 题 9.5

1. 求下列微分方程的通解：

$$(1) y'' - 3y' + 2y = 0; \quad (2) 4y'' + 5y' + y = 0;$$

$$(3) y'' + 6y' + 9y = 0; \quad (4) y'' + 4y' + 5y = 0;$$

$$(5) y'' - y' + 2y = 0; \quad (6) y''' + 2y'' - y' = 0.$$

2. 求下列初值问题的解：

$$(1) \begin{cases} y'' + 2y' + 4y = 0, \\ y(0) = 1, y'(0) = -1; \end{cases} \quad (2) \begin{cases} 4y'' + 4y' + y = 0, \\ y(0) = 0, y'(0) = 2. \end{cases}$$

3. 用待定系数法求下列方程的特解：

$$(1) y'' - 3y' + 5y = 6; \quad (2) y'' + 3y' = 6e^{2x};$$

$$(3) y'' - 9y' + 20y = x + 1; \quad (4) y'' + y = 4\sin x;$$

$$(5) y'' - 3y' + 2y = x \cos x; \quad (6) y'' - 9y = e^{3x} \cos x;$$

$$(7) y'' - y = 2e^x - x^2; \quad (8) y'' + y' = \sin 4x - 2\sin 2x.$$

$$1. (4) \lambda^2 + 4\lambda + 5 = 0, \Delta = 16 - 20 = -4$$

$$\frac{-4 \pm \sqrt{-4}}{2} = -2 \pm i$$

$$C_1 e^{-2 \cos \beta} + C_2 e^{-2 \sin \beta}$$

$$(6) \lambda^3 + 2\lambda^2 - \lambda = 0, \lambda(\lambda^2 + 2\lambda - 1) \quad \lambda_1 = 0, \lambda_{2,3} = \frac{-2 \pm \sqrt{8}}{2} = -1 \pm \sqrt{2}$$

$$\therefore y = C_1 + C_2 e^{-1+\sqrt{2}} + C_3 e^{-1-\sqrt{2}}$$

$$2. (2) 4\lambda^2 + 4\lambda + 1 = 0, (2\lambda + 1)^2 = 0, \lambda_{1,2} = -\frac{1}{2}$$

$$y = (C_1 + C_2 x) e^{-\frac{1}{2}x}$$

$$y' = C_2 e^{-\frac{1}{2}x} + (C_1 + C_2 x) \cdot (-\frac{1}{2}) \cdot e^{-\frac{1}{2}x}$$

$$y(0) = C_1 = 0, \quad y'(0) = C_2 = 2.$$

$$y = 2x e^{-\frac{1}{2}x}$$

$$3. (2) \lambda^2 + 3\lambda = 0, \lambda_1 = 0, \lambda_2 = -3, 2 \text{ 不是特征根}$$

$$Q(x) = a e^{\alpha x}, \text{ 代 } \lambda \text{ 入式}$$

$$(a \alpha^2 + 3a \alpha) \cdot e^{\alpha x} = 6e^{2x}$$

$$\alpha = 2, \quad 10a \cdot e^{2x} = 6e^{2x}, \quad a = \frac{3}{5}$$

$$\text{特解 } y = \frac{3}{5} e^{2x}$$

$$\lambda = \frac{t}{y}, \quad dt = y d\lambda$$

$$f(x, y) = \int_0^{xy} e^{xt^2} dt$$

$$\frac{\partial f(x, y)}{\partial x} =$$

$$\int_0^x \cdot e^{x \cdot \lambda^2 y^2} \cdot y \cdot d\lambda$$

$$= \int_0^x \cdot e^{\lambda^2 x y^2} \cdot y \cdot d\lambda$$

$$\frac{\partial f}{\partial x} = e^{x^3 y^2} \cdot y$$

$$\frac{\partial f}{\partial y} = e^{x^3 y^2} + y \cdot 2x^3 y \cdot e^{x^3 y^2}$$

$$(5) \quad \lambda^2 - 3\lambda + 2 = 0, \quad \lambda_1 = 1, \lambda_2 = 2.$$

$$y = (a_1 x + a_0) \cos x + (b_1 x + b_0) \sin x$$

$$y' = a_1 \cos x - (a_1 x + a_0) \sin x + b_1 \sin x + (b_1 x + b_0) \cos x = (a_1 + b_0 + b_1 x) \cos x + (b_1 - a_1 x - a_0) \sin x$$

$$y'' = -a_1 \sin x - a_1 \sin x - (a_1 x + a_0) \cos x + b_1 \cos x + b_1 \cos x - (b_1 x + b_0) \sin x$$

$$= (-2a_1 - b_1 x - b_0) \sin x + (2b_1 - a_1 x - a_0) \cos x$$

$$(-2a_1 - b_1 x - b_0 - 3b_1 + 3a_1 x + 3a_0 + 2b_1 x + 2b_0) \sin x + (2b_1 - a_1 x - a_0 - 3a_1 - 3b_0 - 3b_1 x + 2a_1 x + 2a_0) \cos x$$

$$= [(b_1 + 3a_1)x + 3a_0 - 2a_1 + b_0 - 3b_1] \sin x + [(a_1 - 3b_1)x + a_0 - 3a_1 + 2b_1 - 3b_0] \cos x$$

$$\begin{cases} b_1 + 3a_1 = 0 \\ 3a_0 - 2a_1 + b_0 - 3b_1 = 0 \\ a_1 - 3b_1 = 1 \\ a_0 - 3a_1 + 2b_1 - 3b_0 = 0 \end{cases} \quad \begin{cases} a_1 = \frac{1}{10} \\ b_1 = -\frac{3}{10} \\ a_0 = -\frac{3}{25} \\ b_0 = \frac{-13}{50} \end{cases}$$

$$\therefore y = \left(\frac{1}{10} x - \frac{3}{25} \right) \cos x + \left(-\frac{3}{10} x - \frac{13}{50} \right) \sin x$$

$$f(x, y) = \int_0^{xy} e^{xt^2} dt$$

$$\frac{\partial f}{\partial x} = xy = z, \quad x = \frac{z}{y}$$

$$f(x, y) = y \cdot e^{x^3 y^2}$$

$$f(x, y) = \int_0^{xy} e^{xt^2} dt$$

$$(8) \quad \lambda^2 + \lambda = 0, \quad \lambda_1 = 0, \lambda_2 = -1$$

$$\textcircled{1} \quad y'' + y' = \sin 4x$$

$$f(x) = A \cos 4x + B \sin 4x$$

$$f'(x) = -4A \sin 4x + 4B \cos 4x$$

$$f''(x) = -16A \cos 4x - 16B \sin 4x$$

$$\therefore (-16B - 4A) \sin 4x + (4B - 16A) \cos 4x = \sin 4x$$

$$\begin{cases} -16B - 4A = 1 \\ 4B - 16A = 0 \end{cases} \quad \begin{cases} A = -\frac{1}{68} \\ B = -\frac{1}{17} \end{cases}$$

$$\therefore f(x) = -\frac{1}{68} \cos 4x - \frac{1}{17} \sin 4x$$

$$\text{故 } y = -\frac{1}{68} \cos 4x - \frac{1}{17} \sin 4x - \frac{1}{5} \cos 2x - \frac{2}{5} \sin 2x$$

$$\textcircled{2} \quad y'' + y' = 2 \sin 2x$$

$$g(x) = A \cos 2x + B \sin 2x$$

$$g'(x) = -2A \sin 2x + 2B \cos 2x$$

$$g''(x) = -4A \cos 2x - 4B \sin 2x$$

$$(2B - 4A) \cos 2x + (-2A - 4B) \sin 2x = 2 \sin 2x$$

$$\begin{cases} 2B - 4A = 0 \\ -2A - 4B = 2 \end{cases} \quad \begin{cases} A = -\frac{1}{5} \\ B = -\frac{2}{5} \end{cases}$$

$$g(x) = -\frac{1}{5} \cos 2x - \frac{2}{5} \sin 2x$$

$$4. \quad y''' + 3y'' + 3y' + y = e^{-x}(x-5)$$

$$(\lambda + 1)^3 = 0, \quad \lambda_{1,2,3} = -1$$

$$\therefore y = x^3 e^{-x} (A_0 x + A_1)$$

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用常数变易法求下列方程的通解：

$$1. \quad y'' + 3y' + 2y = \frac{1}{e^x + 1}.$$

$$2. \quad y'' + y = \frac{1}{\sin x}.$$

$$3. \quad y'' + 4y = 2 \tan x.$$

$$4. \quad y'' + y = 2 \sec^3 x.$$

求下列欧拉方程的通解：

$$5. \quad x^2 y'' - 4xy' + 6y = 0.$$

$$6. \quad x^2 y'' - xy' - 3y = 0.$$

$$7. \quad x^3 y''' + xy' - y = 0.$$

$$8. \quad x^2 y'' + xy' + 4y = 10.$$

$$1. \quad \lambda^2 + 3\lambda + 2 = 0, \quad \lambda_1 = -1, \quad \lambda_2 = -2$$

$$\therefore y = C_1 e^{-x} + C_2 e^{-2x}$$

$$\therefore \text{方程} \begin{cases} C_1'(x) e^{-x} + C_2'(x) e^{-2x} = 0 \\ -C_1'(x) e^{-x} - 2C_2'(x) e^{-2x} = \frac{1}{e^x + 1} \end{cases}$$

$$\text{有} \begin{cases} C_1'(x) = \frac{e^x}{e^x + 1} \\ C_2'(x) = -\frac{e^{2x}}{e^x + 1} \end{cases}$$

$$\therefore C_1(x) = \ln |e^x + 1| + C_1$$

$$C_2(x) = -e^x + \ln |e^x + 1| + C_2$$

$$\therefore y = (\ln |e^x + 1| + C_1) \cdot e^{-x} + (-e^x + \ln |e^x + 1| + C_2) \cdot e^{-2x}$$

$$7. \quad x = e^t, \quad t = \ln x$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \cdot \frac{dy}{dt} = e^{-t} \cdot \frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) \cdot e^{-2t}$$

$$\frac{d^3y}{dx^3} = \left(\frac{d^3y}{dt^3} - 3 \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} \right) \cdot e^{-3t}$$

$$\therefore \text{特征式} = \frac{d^3y}{dt^3} - 3 \frac{d^2y}{dt^2} + 3 \frac{dy}{dt} - y = 0 \quad \lambda^3 - 3\lambda^2 + 3\lambda - 1 = 0, \quad \lambda_{1,2,3} = 1$$

$$\therefore y(t) = (C_1 + C_2 t + C_3 t^2) e^t$$

$$y(x) = [C_1 + C_2 \ln x + C_3 (\ln x)^2] \cdot x$$

9.7.

习 题 9.7

1. 用消元法求解下列微分方程组：

$$(1) \begin{cases} \frac{dx}{dt} = -x - 5y, \\ \frac{dy}{dt} = x + y; \end{cases} \quad (2) \begin{cases} \frac{dx}{dt} = x + y + 2e^t, \\ \frac{dy}{dt} = 4x + y - e^t; \end{cases}$$

1. (2) 由①得 $y = \frac{dx}{dt} - x - 2e^t$, 代入②得

$$\frac{d^2x}{dt^2} - \frac{dx}{dt} - 2e^t = 4x + \frac{dx}{dt} - x - 2e^t - e^t$$

$$\therefore \frac{d^2x}{dt^2} - 2\frac{dx}{dt} - 3x = -e^t, \quad \lambda^2 - 2\lambda - 3 = 0$$

$$\lambda_{1,2} = 3, -1, \text{ 设 } x = Ae^t, \text{ 代入得}$$

$$-4Ae^t = -e^t, \quad A = \frac{1}{4}$$

$$\therefore x = \frac{1}{4}e^t + C_1 e^{3t} + C_2 e^{-t}$$

$$y = 3C_1 e^{3t} - C_2 e^{-t} + \frac{1}{4}e^t - (C_1 e^{3t} + C_2 e^{-t} + \frac{1}{4}e^t) - 2e^t.$$

$$\therefore \begin{cases} x = C_1 e^{3t} + C_2 e^{-t} + \frac{1}{4} e^t \\ y = 2C_1 e^{3t} - 2C_2 e^{-t} - 2e^t \end{cases}$$

$$(3) \begin{cases} \frac{dx}{dt} = 2x - 5y - \sin 2t \\ \frac{dy}{dt} = x - 2y + t, \\ x(0) = 0, y(0) = 1; \end{cases}$$

(3) 由②知 $x=2y+\dot{y}-t$, 代入①知

$$2\dot{y}+\ddot{y}-1=4y+2y-2t-5y-\sin 2t$$

$$\therefore \ddot{y}+\dot{y}=1-2t-\sin 2t, \lambda^2+1=0, \lambda_{1,2}=\pm i$$

(i) 设 $y=A_0 t+A_1$, $\dot{y}=A_0$, $\ddot{y}=0$

$$\therefore A_0 t+A_1=1-2t, A_1=1, A_0=-2$$

(ii) 设 $y=A \sin 2t+B \cos 2t$, $\dot{y}=2A \cos 2t-2B \sin 2t$, $\ddot{y}=-4A \sin 2t-4B \cos 2t$

$$\therefore -3A \sin 2t-3B \cos 2t=-\sin 2t, \begin{cases} A=\frac{1}{3}, \\ B=0 \end{cases}$$

$$\therefore y=C_1 \sin t+C_2 \cos t+\frac{1}{3} \sin 2t-2t+1$$

$$x=2C_1 \sin t+2C_2 \cos t+\frac{2}{3} \sin 2t-4t+2+C_1 \cos t-C_2 \sin t+\frac{2}{3} \cos 2t-2-t$$

$$=(2C_1-C_2) \sin t+(2C_2+C_1) \cos t+\frac{2}{3} \sin 2t+\frac{2}{3} \cos 2t-5t$$

$$\text{代入初始值条件得} \begin{cases} C_1=-\frac{2}{3} \\ C_2=0 \end{cases}$$

$$\therefore \begin{cases} x=-\frac{4}{3} \sin t-\frac{2}{3} \cos t+\frac{2}{3} \sin 2t+\frac{2}{3} \cos 2t-5t \\ y=\frac{2}{3} \sin t+\frac{1}{3} \sin 2t-2t+1 \end{cases}$$

2. 求下列非齐次方程组的一个特解.

$$(1) \begin{cases} \dot{x} = x + 2y, \\ \dot{y} = x - 5\sin t; \end{cases} \quad (2) \begin{cases} \dot{x} = 2x + y + e^t, \\ \dot{y} = -2x + 2t; \end{cases}$$

$$(3) \begin{cases} \dot{x} = 2x - y, \\ \dot{y} = y - 2x + 18t; \end{cases} \quad (4) \begin{cases} \dot{x} = 2x - y, \\ \dot{y} = x + 2e^t. \end{cases}$$

2. (2) 由①得 $y = \dot{x} - 2x - e^t$, 代入②有

$$\ddot{x} - 2\dot{x} - e^t = -2x + 2t$$

$$\ddot{x} - 2\dot{x} + 2x = e^t + 2t, \quad \lambda^2 - 2\lambda + 2 = 0, \quad \lambda_{1,2} = 1 \pm i$$

$$\therefore x = Ae^t + Bt + C$$

$$\dot{x} = Ae^t + B, \quad \ddot{x} = Ae^t$$

$$\text{代入原式, 有 } Ae^t + 2Bt + 2C - 2B = e^t + 2t$$

$$\therefore A = B = C = 1$$

$$\therefore \begin{cases} x = e^t + t + 1 \\ y = -2e^t - 2t - 1 \end{cases}$$